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Local kernel renormalization as a mechanism for feature learning in overparametrized convolutional neural networks

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Empirical evidence shows that fully-connected neural networks in the infinite-width limit (lazy training) eventually outperform their finite-width counterparts in most computer vision tasks; on the other hand, modern architectures with convolutional layers often achieve optimal performances in the finite-width regime. In this work, we present a theoretical framework that provides a rationale for these differences in one-hidden-layer networks; we derive an effective action in the so-called proportional limit for an architecture with one convolutional hidden layer and compare it with the result available for fully-connected networks. Remarkably, we identify a completely different form of kernel renormalization: whereas the kernel of the fully-connected architecture is just globally renormalized by a single scalar parameter, the convolutional kernel undergoes a local renormalization, meaning that the network can select the local components that will contribute to the final prediction in a data-dependent way. This finding highlights a simple mechanism for feature learning that can take place in overparametrized shallow convolutional neural networks, but not in shallow fully-connected architectures or in locally connected neural networks without weight sharing.

Deep learning achieves state-of-the-art performance on a variety of pattern recognition tasks, ranging from computer vision to natural language processing[1]. A minimal theory of deep learning should be able (at least) to predict practically relevant scores, such as the training and test accuracy, from knowledge of the training data. On a fundamental level, this means addressing the so-called feature learning regime[2,3], i.e., the capability of deep networks to automatically detect useful representations from raw data[4].

Researchers have investigated numerous strategies to address these challenges, including (i) large-width (infinite-width) limit of neural networks, that is the regime where the size of the hidden layers N_ℓ (with $\ell = 1 \dots L$, L being the network depth) is taken to infinity keeping finite the number of examples P in the trainset. This establishes the first and only analytically solvable model of deep architectures, unraveling a rich phenomenology that ranges from the so-called lazy training[5–16], to the feature learning regime[17–26]; (ii) perturbative corrections to the infinite-width limit[27–31] (iii) deep linear models[32–35]. Despite their general interest, these results do not describe neural networks in realistic regimes: the infinite-width approximation blatantly departs from neural networks practice, where the number of training examples P is comparable in magnitude, or even larger than the layer widths, while the assumption of linear activation function dramatically reduces the expressivity of the models.

In view of describing verisimilar settings of learning, progress has been made in the context of Bayesian neural networks in the so-called proportional regime, defined as the thermodynamic limit where the size of the hidden layers and of the training set are taken to infinity while their ratio $\alpha_\ell = P/N_\ell$ is kept finite. In this framework, one can build

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a thermodynamic descriptions based on scale separations[36,37], or directly extract an "effective action" whose minimization describes the entire effect of training in this regime[38–40]. The Bayesian framework for deep neural networks (DNNs) was pioneered in the classic reference[41], and its connection with equilibrium statistical mechanics has recently been reviewed in ref. 42. A standard introductory reference to Gaussian Processes, which are fundamental both in the infinite-width and proportional limit, is the book[43].

The picture that emerges from the aforementioned theoretical results is completed by a series of recent empirical observations that highlight how infinite-width performances are state-of-the-art in some contexts. For instance, the large-scale empirical study conducted by the authors of[44], where they explore a plenitude of possible settings to improve generalization in finite and infinite-width models, prompts a few striking observations: infinite-width kernels systematically outperform their finite-width counterpart in the case of fully-connected (FC) deep neural networks; convolutional neural networks (CNNs) with finite number of channels often outperform their corresponding infinite-width kernel performance. Several similar observations have been reported in the literature: the authors of ref. 45 very recently pointed out that infinite-width deep FC neural networks eventually outperform their finite-width counterpart as the size of the training set grows. In one of the seminal papers dealing with the infinite-width limit[8], the authors observe that increasing the hidden layers size leads to optimal test accuracy on deep FC architectures trained on MNIST and on CIFAR10, two of the benchmark datasets for computer vision learning problems. In ref. 10 this observation is extended to locally connected networks without weight sharing (LCNs). Interestingly, the authors of refs. 46,47 observed that FC networks with mean field scaling (that display feature learning in the infinite-width limit) are systematically outperformed by infinite-width kernels.

These findings suggest that CNNs leverage a better feature-learning mechanism at finite width than FC and LCNs, and prompt at least two conceptual questions: (i) why is it ultimately convenient to employ large-width architectures when only FC layers are available? (ii) Why is this not the case when convolutional layers are employed? And how does a CNN operatively exploit the finite-width regime for efficient feature learning?

In this work, we try to rationalize the aforementioned empirical observations through the lens of kernel renormalization, that is a physical consequence of the proportional-width description, identified in ref. 38 for non-linear networks. First, we derive an effective action for two simple shallow models (CNN and LCN), showing that its functional form is shared among architectures and related to the FC case. Second, we show that the theoretical framework of ref. 38 provides a simple, yet very instructive answer to question (i): the performance of a proportional-width shallow network can be obtained with the corresponding infinite-width kernel and a suitable choice of the Gaussian priors over the weights of each layer. Finally, we compare the result obtained for CNNs with the one available for FC networks. We find a striking difference in the way the kernel of the two architectures renormalize at proportional width: whereas the FC kernel is just globally renormalized by a scalar parameter, the CNN kernel undergoes a local renormalization, meaning that many more free parameters are allowed to be fine-tuned during training (see also Fig. 1). We employ this finding to provide preliminary insight to question (ii): we highlight a simple mechanism for feature learning that can take place in finite-width shallow CNNs, but neither in shallow FC architectures nor in LCNs without weight sharing.

Results

Functional form of the effective action for learning in the proportional regime is shared among shallow architectures with local connections

To address the questions raised in the introduction of this manuscript, we analytically investigate two types of one-hidden-layer (1HL) architectures with local connections, namely CNNs and LCNs with a FC readout layer, whose explicit definitions are given in the Methods section of this manuscript.

The two fundamental ingredients of a convolutional layer are local connectivity and weight sharing: whereas a given neuron in a FC hidden layer receives input from all the neurons in the previous layer, neurons in a CNN are arranged in a d-dimensional array that reflects the corresponding d-dimensional arrangement of the input data (e.g., $d = 2$ for images). Each neuron in a given layer here interacts only with a local neighborhood of neurons in the previous layer, in a translational invar-

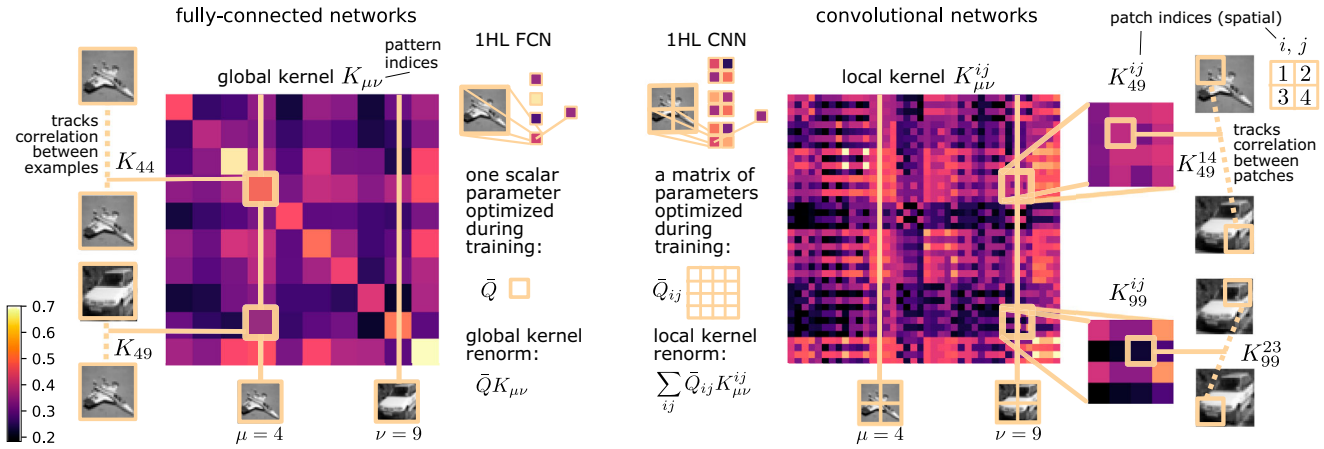


Fig. 1 | Global and local kernels in FC and convolutional one hidden layer networks. Our calculations in the proportional limit $P, N_1 \to \infty$ at fixed $\alpha_1 = P/N_1$ reveal a striking difference between FC and CNNs. In the first case, the network performances are linked to the NNGP kernel $K_{\mu\nu}$, which is only capable of tracking global correlations between different training patterns (μ, ν) (left). A FC network in this regime will use the information contained in the dataset to fine-tune a single parameter, $\bar{Q}$, that will globally rescale the kernel matrix $K_{\mu\nu}$. On the other hand, in the CNN case, we identify a local kernel with two additional (spatial) indices. If we consider $2d$ images, the effect of the local components is to track local correlations between different patches i and j of (possibly) different pairs of images (μ, ν) (right). The spatial information contained in these correlations allows the CNN to optimize a matrix of parameters $\bar{Q}_{ij}$, which will renormalize the kernel matrix $K^{ij}_{\mu\nu}$ in a non-trivial way. In the figure, we display the four-index local kernel $K^{ij}_{\mu\nu}$ as a matrix using the multi-index notation $K_{(\mu,\, i),(\nu,\, j)}$ and choosing an ordering for the multi-index (μ, i). Global and local kernels are computed for a subset of $P = 10$ gray-scaled CIFAR10 images down-sized to $N_0 = 784$. The local kernels are empirically evaluated for $2d$ convolutions with linear size of the filter mask $M = 14$ and stride $S = 14$ (non-overlapping filters).

iant way implemented via a shared (usually small) d-dimensional mask of learnable weights. These operations define a single convolutional channel that takes as input a d-dimensional array and outputs another d-dimensional array, whose dimensions are determined by technical details (such as the stride, padding and dimension of the filter mask). Usually, many of these channels are piled up in a convolutional layer to form, overall, a $(d+1)$-dimensional array. The LCN represents a variation of a CNN, where there is local connectivity but not weight sharing.

Our goal is to derive an effective action in the thermodynamic proportional limit where the number of convolutional channels N_c and the dataset size P are taken to infinity with their ratio $\alpha_c = P/N_c$ finite. We do so by computing the partition function of the two architectures at temperature $T = 1/\beta$, that is:

$$Z = \int \mathrm{d}\mu(\theta)\, \exp\{-\beta \mathcal{L}(\theta)\}\,, \tag{1}$$

where θ represents the collection of the network's weights, $\mathcal{L}$ is the mean squared error loss function defined in the Methods, and the notation $\mathrm{d}\mu(\theta)$ indicates collective integration over the weights and their prior probability distribution $\mu(\theta)$. The explicit definitions of the functions implemented by CNNs and LCNs can be found in the Methods. As we explicitly show in the Supplementary Methods 1 and 2[48], we find that Z can be approximated as an integral over an ensemble of positive semi-definite matrices Q and $\bar{Q}$: $Z = \int \mathcal{D}Q\mathcal{D}\bar{Q} e^{-N_c S(Q,\bar{Q})/2}$, where the effective action is given by:

$$\begin{aligned} S(Q,\bar{Q}) \equiv \; & -\mathrm{Tr}\, Q\bar{Q} + \mathrm{Tr}\log(\mathbb{1}+Q) \\ & + \tfrac{\alpha_c}{P}\mathrm{Tr}\log\beta\left(\tfrac{1}{\beta} + K^{(\mathrm{R})}(X,X;\bar{Q})\right) \\ & + \tfrac{\alpha_c}{P}\mathbf{y}^{\top}\left(\tfrac{1}{\beta} + K^{(\mathrm{R})}(X,X;\bar{Q})\right)^{-1}\mathbf{y}\,. \end{aligned} \tag{2}$$

The renormalized kernel $K^{(\mathrm{R})}(X,X;\bar{Q})$ is a $P\times P$ matrix that depends on the specific architecture (CNN or LCN), the order parameters in $\bar{Q}$, and incorporates the dependence of the action on the trainset $X = \{x^\mu\}_{\mu=1}^P$. The P-dimensional vector $\mathbf{y} = \{y^\mu\}_{\mu=1}^P$ comprises all trainset labels. The trace in the first two terms is over $\lfloor N_0/S\rfloor \times \lfloor N_0/S\rfloor$ operators, where N_0 is the input dimension, S is the stride of the convolutional filters and $\lfloor x\rfloor$ indicates the largest integer less than or equal to x (see Methods section for more details). The trace in the third term and the scalar products with the output labels $\mathbf{y}$ lie in a P-dimensional vectorial space. Here and in the following we will imply that all the kernel functions $K(\cdot,\cdot)$ and renormalized ones $K^{(\mathrm{R})}(\cdot,\cdot;\bar{Q})$ are two-point functions, using the following shorthand notation to denote matrices and vectors:

$$K(X,X):\ [K(X,X)]_{\mu\nu} = K(x^\mu,x^\nu)\,, \tag{3}$$

$$K(x',X):\ [K(x',X)]_{\mu} = K(x',x^\mu)\,. \tag{4}$$

The expressions of the renormalized kernels, respectively for CNNs and LCNs, read:

$$K^{(\mathrm{R})}_{\mathrm{CNN}}(x^\mu,x^\nu;\bar{Q}) = \frac{1}{\lambda_1\lfloor N_0/S\rfloor}\sum_{ij=1}^{\lfloor N_0/S\rfloor}\bar{Q}_{ij}K^{ij}(x^\mu,x^\nu)\,, \tag{5}$$

$$K^{(\mathrm{R})}_{\mathrm{LCN}}(x^\mu,x^\nu;\bar{Q}) = \frac{1}{\lambda_1\lfloor N_0/S\rfloor}\sum_{i=1}^{\lfloor N_0/S\rfloor}\bar{Q}_{i}K^{ii}(x^\mu,x^\nu)\,, \tag{6}$$

where λ_1 is the Gaussian prior of the readout layer. The local kernel $K^{ij}_{\mu\nu} \equiv K^{ij}(x^\mu,x^\nu)$, named local due to the dependence on the i,j indices that run over the input coordinates, is given in terms of the elements of a local covariance matrix $C^{ij}_{\mu\nu}$ as follows:

$$C^{ij}_{\mu\nu} = \frac{1}{\lambda_0 M}\sum_{m=-\lfloor M/2\rfloor}^{\lfloor M/2\rfloor} x^\mu_{Si+m}x^\nu_{Sj+m}, \tag{7}$$

$$\begin{aligned} K^{ij}_{\mu\nu} &= \int dt_1\, dt_2\, \mathcal{N}_{\boldsymbol{t}}\left(\boldsymbol{0},\tilde{C}^{ij}_{\mu\nu}\right)\sigma(t_1)\sigma(t_2)\,, \\ \tilde{C}^{ij}_{\mu\nu} &= \begin{pmatrix} C^{ii}_{\mu\mu} & C^{ij}_{\mu\nu} \\ C^{ij}_{\mu\nu} & C^{jj}_{\nu\nu}\,. \end{pmatrix}. \end{aligned} \tag{8}$$

where λ_0 is the hidden layer Gaussian prior, the index m runs over the convolutional mask M, and S is the stride. We use the notation $\mathcal{N}_{\boldsymbol{t}}(\boldsymbol{0},\tilde{C}^{ij}_{\mu\nu})$ to indicate a normalized Gaussian distribution of the vector $\boldsymbol{t} = (t_1, t_2)$ with zero mean and covariance matrix $\tilde{C}^{ij}_{\mu\nu}$.

Such a local kernel has already been found in the seminal work on the Gaussian process limit of infinite-width CNNs[10,11]. In the limit of infinitely many channels $N_c \gg P$, one finds that CNNs and LCNs evolve according to the following averaged kernel:

$$\bar{K}_{\mu\nu} = \frac{1}{\lambda_1\lfloor N_0/S\rfloor}\sum_{i=1}^{\lfloor N_0/S\rfloor}K^{ii}_{\mu\nu}\,, \tag{9}$$

which defines the infinite-width performance of locally-connected models. In our framework, we recover this result by solving the saddle-point equations for the matrix $\bar{Q}$ in the limit of $\alpha_c \to 0$, where one finds $\bar{Q}_{ij} = \delta_{ij}$ and only the diagonal components of the local kernel contribute to the prediction. Note that the effective action that describes FC networks, originally derived in ref. 38 and reported in Eq. (21) of the Methods Section, is recovered setting $S = N_0$ in Eq. (2).

The shared functional form of the effective action induces the same Gaussian statistics of the predictor for an unseen example (x^0, y^0) in all three architectures (FC, LCNs, CNNs), namely:

$$\begin{aligned} & p(f^0|X,\mathbf{y}) = \mathcal{N}(\Gamma,\sigma^2) \\ & \Gamma \equiv \left\langle f^0\right\rangle = \left(K^{(\mathrm{R})}(x^0,X)\right)^T\left(\tfrac{1}{\beta}+K^{(\mathrm{R})}(X,X)\right)^{-1}\mathbf{y} \end{aligned} \tag{10}$$

$$\begin{aligned} \sigma^2 \equiv & \left\langle (f^0)^2\right\rangle - \left\langle f^0\right\rangle^2 = K^{(\mathrm{R})}(x^0,x^0) + \\ & -\left(K^{(\mathrm{R})}(x^0,X)\right)^T\left(\tfrac{1}{\beta}+K^{(\mathrm{R})}(X,X)\right)^{-1}K^{(\mathrm{R})}(x^0,X)\,, \end{aligned} \tag{11}$$

where $f^0 \equiv f_{1\mathrm{HL}}(x^0)$ is the output of the network corresponding to input x^0, and the angular brackets represent the average over the equilibrium Gibbs distribution $p_\beta(\theta) = e^{-\beta\mathcal{L}(\theta)}\mu(\theta)/Z$. An explicit derivation of this posterior statistics can be found in ref. 38. The generalization error over the same new example is computed from the bias and variance of the predictor:

$$\left\langle \epsilon_{\mathrm{g}}(x^0,y^0)\right\rangle \overset{\mathrm{def}}{=} \left\langle (f^0-y^0)^2\right\rangle = (y^0-\Gamma)^2+\sigma^2. \tag{12}$$

Tuning Gaussian priors in infinite-width 1HL FC networks is equivalent to training in the proportional regime

A physical consequence of the theoretical framework presented in ref. 38 (and briefly summarised in the Methods section) stems from inspection of the network's predictive probability distribution, given in Eq. (12), that holds for FC 1HL networks with the following renormalized Kernel:

$$K^{(\mathrm{R})}_{\mathrm{FC}}(x,x') = \frac{\bar{Q}}{\lambda_1}K(x,x') \tag{13}$$

where $K(x,x')$ is the Neural Network Gaussian Process (NNGP) kernel, found in the context of Bayesian infinite-width[8,9], that plays the same role in FC as Eq. (8) in locally-connected networks.

First, we notice that the renormalized kernels $K^{(R)}_{FC}$ enter the predictor's statistics for the unseen test element exactly in the same way the NNGP kernel does in the infinite-width limit, and thus it completely determines the generalization performance at proportional width, once the saddle-point equations for the parameter $\bar{Q}$ are solved[38].

Second, we observe that the scalar parameter $\bar{Q}$ always appears in combination with the corresponding Gaussian prior λ_1 as $\bar{Q}/\lambda_1$, meaning that, once evaluated on the saddle-point, $\bar{Q}$ is just a data-dependent scalar renormalization of the Gaussian prior. This implies that, once the size of the training set and the activation function σ are fixed, it is always possible to re-obtain the performance of any finite-width network just by carefully fine-tuning the Gaussian prior λ_1 in the corresponding infinite-width kernel. Therefore, the generalization performance of any finite-width one hidden layer FC network is bounded by the one of a suitable infinite-width kernel with optimal choice of the Gaussian prior. Note that the aforementioned argument can be generalized to networks with depth $L>1$, using the explicit expression for the renormalized NNGP kernel found in ref. 38 and noting that the Q_ℓ's at each layer ℓ are in one-to-one correspondence to the Gaussian priors λ_ℓ at the same layer.

The empirical observation that infinite-width one hidden layer FC networks seem to systematically outperform their finite-width counterpart –which now finds a possible explanation in this framework– points to the somewhat disappointing consequence that feature learning is not particularly effective in networks with FC layers alone at proportional width.

We now move to the second question raised in the Introduction of this manuscript. How –contrarily to what occurs in deep nets with FC layers only– do deep architectures with convolutional layers exploit the finite-width regime for efficient feature learning? In view of the previous discussion, one possibility is that CNNs at proportional width could be able, in some way, to break the global (trivial) renormalization of the infinite-width kernel that occurs in FC deep networks. This will be the topic under consideration in the next section.

A simple mechanism for feature learning in proportional-width CNNs: local kernel renormalization

The precise form of kernel renormalization exhibited in Eq. (5) is what we define as local kernel renormalization.

Our claim is that the matrix $\bar{Q}$ provides a compact description of feature learning in the CNN model under consideration: the feature matrix elements $\bar{Q}_{ij}$ that are optimized during training are in fact in one-to-one correspondence with the combinations of pairs of the $\lfloor N_0/S \rfloor$ patches of the local covariance matrix, that are defined in turn by patches of the trainset elements x^μ. From Eq. (5), we can now interpret the matrix $\bar{Q}$ as a feature, data-dependent matrix that tells us how much a given component of the local kernel contribute to the renormalized kernel $K^{(R)}_{CNN}$.

Note that the feature matrix elements Q_{ij} have also a natural physical interpretation as the thermal average of the readout vectors correlations corresponding to patches (i,j). We show this explicitly in the Supplementary Methods 3[48].

The local kernel renormalization that takes place in this CNN model in the proportional limit is completely different from the trivial global one that occurs in the corresponding FC one hidden layer network. At this point one can wonder whether both the ingredients in CNNs design, local connectivity and weight sharing, are needed to observe the phenomenon of local kernel renormalization. The answer is affirmative: local kernel renormalization does not occur in a shallow network with local connectivity only, as we explicitly check in the Supplementary Methods 1[48]. In that case, only the diagonal components of the local kernel enter the effective action, as shown in Eq. (6), therefore the network is not sensible to spatial correlations between different patches (see also Table 1 for a summary of the analytical results).

Table 1 | Summary of the analytical results for one hidden layer networks

Architecture	Kernel type	Ifninite-width kernel	Renormalized kernel
FC	$K_{\mu\nu}$	$K_{\mu\nu}$	$\bar{Q}K_{\mu\nu}$
LCN	$K^{ii}_{\mu\nu}$	$\sum_i K^{ii}_{\mu\nu}$	$\sum_i \bar{Q}_i K^{ii}_{\mu\nu}$
CNN	$K^{ij}_{\mu\nu}$	$\sum_i K^{ii}_{\mu\nu}$	$\sum_{ij} \bar{Q}_{ij} K^{ij}_{\mu\nu}$

We identify three different types of kernels in the Bayesian effective action for FC, LC and convolutional networks: a global kernel in the FC case, a local diagonal kernel in LCNs and a local (with additional non-diagonal components) in CNNs. The infinite-width kernel that describes LCNs and CNNs is the same, as shown in refs. 10,11. In the proportional limit, the renormalization of the FC kernel is scalar, i.e., the NNGP kernel is only globally rescaled by a data-dependent number $\bar{Q}$. In LCNs, we can adjust a vector of components $\bar{Q}_i$, which determines how much the self-correlations of the i-th patch of the diagonal local kernel do contribute to the final prediction. In CNNs, due to locality and weight-sharing, we can fine-tune a full matrix $\bar{Q}_{ij}$, which controls how much the local correlations of patches i and j will enter the effective kernel learned by the network after training. This is what we call local kernel renormalization.

To illustrate the possible practical benefits of the local kernel renormalization mechanism predicted by our effective theory, it is useful to consider the bias-dominated regime, i.e. the limit where the variance term in Eq. (12) shrinks to zero (note that this can be obtained in practice by ensembling over a large number of different realizations of the trained network).

Let us fix $\lambda_0 = 1$ without loss of generality (changing λ_0 amounts to a global rescaling of the single components of the training patterns). In the bias-dominated regime, the generalization loss of an infinite-width CNN with NNGP kernel $\bar{K}$ given by Eq. (9) (corresponding to the choice of $\bar{Q}^* = \mathbb{1}$ in Eq. (5)) is lower bounded looking for the optimal values of two parameters: the temperature T and Gaussian prior over the last layer λ_1.

The result of this optimization for a supervised learning problem using a subset of the CIFAR10 dataset (see Supplementary Methods 5 for details[48]) is shown in Fig. 2 for different values of the training set size P (purple dots). Starting from the optimal infinite-width values T^* and λ_1^*, we can explicitly check that local kernel renormalization, i.e., the choice of a feature matrix $Q \neq \mathbb{1}$ can indeed further improve the generalization performance (see orange markers in Fig. 2 and two examples of matrices Q_{ij} found using the numerical protocol outlined in Supplementary Methods 5[48]). The code used to perform these experiments is available as an open-source package in ref. 49.

It is worth stressing that this is just a proof-of-principle analysis, and we are currently working on the accurate comparison between our effective theory and Bayesian experiments based on Langevin dynamics, in the same spirit of what we have recently done for fully-connected architectures in ref. 39. We conducted preliminary analysis with Bayesian CNNs with 2d convolutional filters, which we report in the Supplementary Discussion[48].

Empirical evidence for global and local kernel renormalization

Extracting the renormalized kernel from the measure of a physical observable is not straightforward, and we have no way, at the moment, to directly access it in numerical experiments. One indirect experimental blueprint of the form of kernel renormalization at proportional width in different architectures is given by the similarity matrix of the internal representations before and after training. We define this

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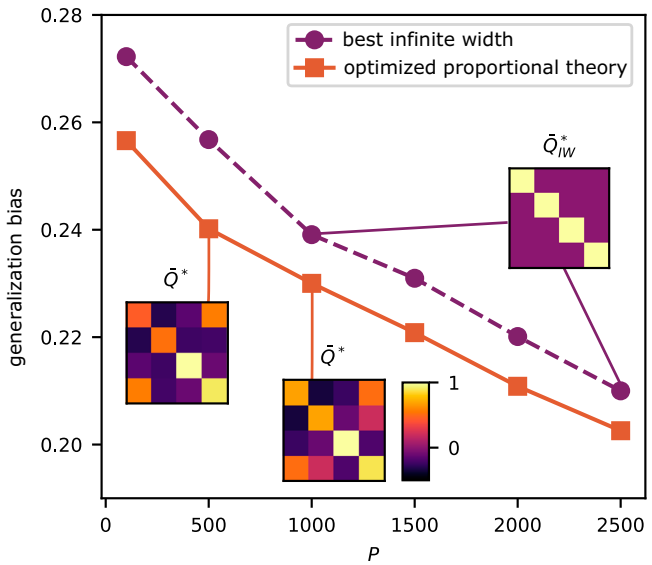


Fig. 2 | Local kernel renormalization can in principle outperform the best infinite-width generalization performance. Predicted generalization bias as a function of the number of training examples P. (dots) Best infinite-width performance, corresponding to the solution $\bar{Q}^*_{\rm IW} = \mathbb{1}$. For each P, the optimal values for the hyperparameters $\lambda^*_{1,P}$ and T^*_P are obtained with a grid search in the ranges $\lambda_1 = [0.1, 10]$, $T = [0.02, 0.001]$. (squares) Generalization bias of a proportional network with the optimal infinite-width hyperparameters $\lambda^*_{1,P}$, T^*_P, and with $\bar{Q}^* \neq \mathbb{1}$, depicted for $P = 500$ and $P = 1000$. The optimal values $\bar{Q}^*$ are obtained for each P with a gradient-based procedure illustrated in Supplementary Methods 5[48]. The binary task under consideration is constructed with the classes "cars" and "planes" of the gray-scaled CIFAR10 dataset (see Supplementary Methods 5 for more details[48]). The open-source package to reproduce these experiments can be found in ref. 49.

observable for both FC and CNNs as:

$$O_{\mu\nu} \equiv \frac{1}{N_1}\sum_{i=1}^{N_1} \sigma(h_i^\mu)\sigma(h_i^\nu)\,, \tag{14}$$

where N_1 denotes the number of neurons in the last layer (for CNNs we have $N_1 = N_c\lfloor N_0/S \rfloor$). Note that the observable in Eq. (14) coincides with the empirical correlation of the post-activations when the activation function is odd.

We can track the effect of training by taking the difference between the similarity matrix at initialization, which is by definition the NNGP kernel (up to the Gaussian prior λ_1), and the same quantity after training: $\Delta K_{\mu\nu} \equiv \langle O_{\mu\nu}\rangle - K_{\mu\nu}$, where the average is done over the Gibbs ensemble of the weights. We analytically compute this observable for FC and CNNs, as shown in Supplementary Methods 4[48]. The final result in the two cases respectively reads (at zero temperature):

$$\Delta K^{\rm FC}_{\mu\nu} = -\frac{1}{N_1}\left[K_{\mu\nu} - \frac{\lambda_1}{\bar{Q}} y^\mu y^\nu\right]; \tag{15}$$

$$\Delta K^{\rm CNN}_{\mu\nu} = -\frac{1}{\lambda_1 N_1 \lfloor N_0/S\rfloor}\sum_{ij,\lambda\rho=1}^{\lfloor N_0/S\rfloor,P} \bar{Q}_{ij} P^{ij}_{\mu\lambda\nu\rho} \times\left[\left(K^{(\rm R)}_{\rm CNN}\right)^{-1}_{\lambda\rho} - \sum_{\epsilon\omega=1}^{P}\left(K^{(\rm R)}_{\rm CNN}\right)^{-1}_{\lambda\epsilon}\left(K^{(\rm R)}_{\rm CNN}\right)^{-1}_{\rho\omega} y^\epsilon y^\omega\right], \tag{16}$$

where $P^{ij}_{\mu\lambda\nu\rho} \equiv \frac{1}{2}\sum_k \left[K^{ik}_{\mu\lambda}K^{kj}_{\nu\rho} + K^{ki}_{\mu\lambda}K^{jk}_{\nu\rho}\right]$. Let us now highlight a few interesting physical implications that directly follow from these formulas and can be checked with numerical experiments:

(i) the difference between the trained and untrained similarity matrix $\Delta K^{\rm FC}_{\mu\nu}$ converges to zero in the thermodynamic limit $P, N_1 \to \infty$ at fixed $\alpha_1 = P/N_1$, as long as the terms in the square brackets are of order $o(N_1)$. This means that in the proportional regime under consideration the effect of training on the internal representations is just a finite-size correction $O(1/N_1)$ and the trained similarity matrix stays very close to its value at initialization. In particular, we expect that the empirical distribution of the elements of the matrix $\Delta K^{\rm FC}_{\mu\nu}$ will be increasingly peaked around zero in finite-size scaling experiments where we consider increasing values of P and N_1 keeping their ratio $\alpha_1 = P/N_1$ fixed;

(ii) by choosing a learning task with labels $y^\mu = 0, 1$ and input patterns x^μ such that the matrix elements of the kernel $K_{\mu\nu}$ are on average centered in zero, the distribution of the matrix elements of the block $\Delta K^{\rm FC}_{\mu\nu}|_{11}$ (corresponding to the subset of training patterns with label one) should drift towards zero at a rate $1/N_1$, if one performs experiments at constant α_1, while increasing P and N_1. Interestingly, the remaining blocks of the matrix $\Delta K^{\rm FC}_{\mu\nu}$ should not be affected by this drift;

(iii) the difference between the trained and untrained convolutional similarity matrix $\Delta K^{\rm CNN}_{\mu\nu}$ displays a different behavior, due to local kernel renormalization. For instance, even if we restrict our analysis to the learning setting outlined in bullet (ii), the elements of the block $\Delta K^{\rm CNN}_{\mu\nu}|_{01}$ should not identically converge to zero in the thermodynamic proportional limit. A straightforward scaling analysis in fact shows that the contribution due to the term in the square bracket does not vanish in the proportional limit: if P is the same order of magnitude of N_1, the double sum of P O(1) kernels on the rhs of Eq. (16) scales as P and it compensates the vanishing prefactor $1/N_1$, leading to a finite $\Delta K^{\rm CNN} \propto O(P/N_1)$.

In Fig. 3 we present the results of a finite-size scaling analysis to check the aforementioned predictions, where we trained one hidden layer FC and CNNs (1d convolutions) on a synthetic dataset composed of random Gaussian patterns whose labels are given by a linear teacher function, i.e. $y = \frac{1}{2}[1 + \mathrm{sign}(t \cdot x)]$, where t is an N_0-dimensional vector with unitary entries. Overall, the fact that the change in the internal representations of FC networks is a finite-size effect is confirmed by the experiments. On the other hand, the same numerical simulations with CNNs clearly identify a genuine change in the internal representations that occurs also in the thermodynamic proportional limit. We note that such a clean signature of the differences between FC and CNNs derive also from our choice of working close to zero temperature (where the architectures can precisely fit the training data). Equations at finite T are more involved, and measuring the indirect effects of global and local kernel renormalization in this case is more difficult. Interestingly, even though our theory should be valid only in the Bayesian setting, it turns out to be also predictive for a learning algorithm that does not explicitly sample from the Gibbs posterior distribution: the aforementioned experiments were carried out using gradient descent and the so-called Adam optimizer[50], with the code available at ref. 51. In the Supplementary Methods 5[48], we provide more details of these experiments, and additional results for $\alpha = 0.1, 10$ (see Supplementary Figs. 1 and 2, Supplementary Table 1).

Discussion

In this work we have derived an effective action in the proportional limit for CNNs and LCNs with one hidden layer, and we have extensively compared these results with the FC case found in[38].

We have highlighted a striking difference in the kernel renormalization mechanism that the three different architectures undergo at proportional width, summarized in Table 1, and we have suggested

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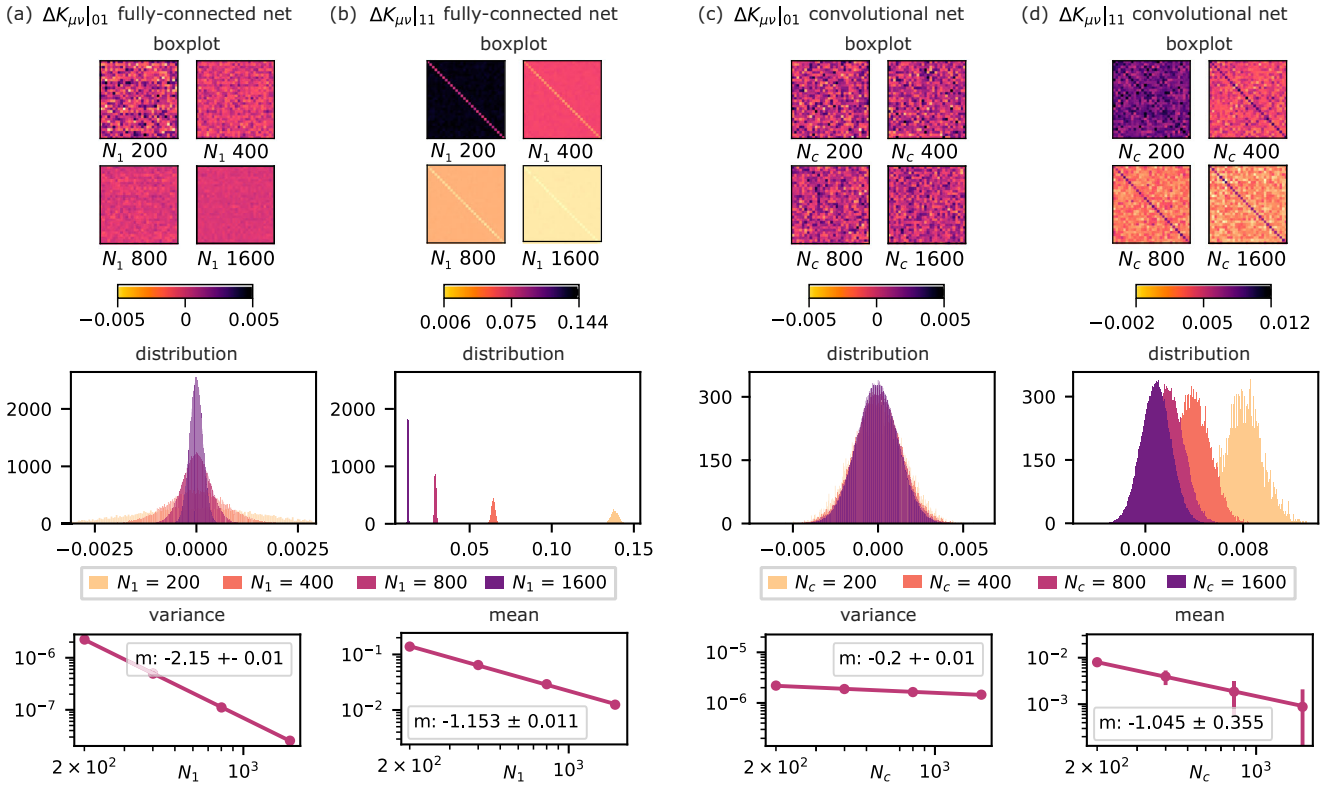


Fig. 3 | Predicting the effect of global and local kernel renormalization on the internal representations of FC and CNNs. A finite-size scaling analysis. We show the results of numerical experiments performed to check scaling predictions of Eqs. (15), (16) for the difference $\Delta K_{\mu\nu}$ between the similarity matrix after and before training. In particular, we train architectures with a single hidden FC/1*d*-convolutional layer with tanh activation (see details in Supplementary Methods 5[48]) on a learning task with random Gaussian inputs and zero/one labels given by a linear teacher. We perform a finite-size scaling analysis by keeping $\alpha_1 = 1$ and choosing increasing values of P, N_1. In panel (**a**) we observe that the variance of the empirical distribution of the matrix elements of the zero-one block $\Delta K^{\mathrm{FC}}_{\mu\nu}|_{01}$ converges to zero as N_1 grows and α_1 is kept fixed to one, in agreement with the prediction of Eq. (15) (middle). In particular, the variance goes to zero as $1/N_1^2$ (bottom). The fact that the similarity matrices after and before training are more and more similar in the thermodynamic limit is also visible in the array plot of a subsample of the zero-one block $\Delta K^{\mathrm{FC}}_{\mu\nu}|_{01}$ (top). In panel (**b**) we display the histogram of the distribution of the matrix elements of the one-one block $\Delta K^{\mathrm{FC}}_{\mu\nu}|_{11}$ (middle). As predicted by Eq. (15), the distribution shrinks while drifting towards zero at a rate $\sim 1/N_1$ (bottom). Overall, (**a**, **b**) represent an indication that the change in the after-training internal representations in 1HL FC networks is a finite-size effect. In (**c**, **d**) we repeat the same experiment for an architecture with a 1*d*-CNN. The striking difference wrt the FC case is that the variance of the empirical distribution of the matrix elements $\Delta K^{\mathrm{CNN}}_{\mu\nu}$ does not shrink to zero as N_c grows and α_c is kept fixed to one, as one can also appreciate looking at the array plot of a subsample of the block matrices $\Delta K^{\mathrm{CNN}}_{\mu\nu}|_{01}$ and $\Delta K^{\mathrm{CNN}}_{\mu\nu}|_{11}$. In all bottom panels, m is the slope of the displayed line, while error bars represent one standard deviation of the matrix entries, averaged over different initializations.

a mechanism for feature learning (local kernel renormalization) that can take place in CNNs, but not in architectures with LC or FC layers alone.

A technical requirement of our results is a Gaussian equivalence, informally justified using a general class of central limit theorems (such as the Breuer-Major theorem in ref. 38). This suggests that the dimension of the input N_0 should scale to infinity at the same rate as the size of the dataset P. If this is the case, as it was in ref. 38, the stride S should be chosen to scale extensively in N_0, so that the final effective action will depend on a finite number of order parameters. For the sake of completeness, we performed numerical analysis of the probability distribution involved in the Gaussian approximation that we show in Supplementary Fig. 3[48].

We have made a preliminary study on the effect of global and local kernel renormalization at the level of the internal representations of trained networks, which is in agreement with our predictions. Additionally, we have shown that the local kernel renormalization mechanism is sufficient for the CNN to outperform its best infinite-width kernel, by finding near-optimal matrices $\bar{Q}^* \neq \mathbb{1}$ that produce a lower generalization error than the infinite-width solution $\bar{Q}^*_{\mathrm{IW}} = \mathbb{1}$ with the same choice of hyperparameters.

It is worth stressing here that the aforementioned observations are not ruling out at all other possible forms of feature learning in FC architectures: (i) Renormalization of the infinite-width kernel is not the only source of feature learning possible. Higher order kernels, irrelevant in the infinite-width limit, may play a role in feature learning, especially as long as one considers the case where the size of the dataset P roughly scales as the number of parameters $\sim L \times N^2$ ($N_\ell = N \ \forall \ \ell$) of the FC deep neural network. The recent work[52] is a notable example of how a FC network can learn a convolutional structure in special settings; (ii) Our result holds for Bayesian learning, i.e. when the weights of the networks are sampled from the canonical Gibbs ensemble. This is only obtained if training is governed by a Markov chain Monte Carlo. Practical learning algorithms with state-of-the-art optimizers may behave differently and possibly activate alternative forms of feature learning; (iii) Our analysis holds in the standard parametrization setting where the last layer is normalized as $1/\sqrt{N_1}$, and we cannot describe the mechanism for feature learning that may occur in the mean field regime[17] where the last layer is normalized as $1/N_1$. Nonetheless, it must be also said that the empirical evidence provided in refs. 8,10,11,44,45,47 is quite against the fact that these forms of feature learning can play a significant role in improving the generalization performance of FC networks.

We conclude by pointing out a number of possible follow-ups of this work: (i) It would be interesting to extend our framework to L convolutional layers, following the same strategy proposed for FC

layers in[38]; (ii) Our current approach cannot cover the mean-field regime, where the feature learning mechanism for FC and CNNs may quantitatively change. We think that the formalism presented in[36] might be more suited to investigate finite-width effects in this case; (iii) We are writing a detailed open-source numerical routine, already publicly available online[49], to extensively check the theoretical predictions of our effective theory for FC, LC and CNNs. This step will also serve to point out the limitations of the present framework.

Methods

Setting of the learning problem and recap of results for fully-connected networks

We consider a supervised regression problem with training set $X = \{x^\mu\}_{\mu=1}^P$, where each $x^\mu \in \mathbb{R}^{N_0}$, with corresponding labels $\mathbf{y} = \{y^\mu\}_{\mu=1}^P$ with $y^\mu \in \mathbb{R}$. We also restrict our analysis to quadratic loss function $\mathcal{L}$. For a generic one hidden layer network parameterized by the weights $\theta = (W, v)$ that implements the function $f_{1\text{HL}}(x;\theta)$, we define the loss function as:

$$\mathcal{L} = \frac{1}{2}\sum_{\mu=1}^{P}\left[y^\mu - f_{1\text{HL}}(x^\mu;\theta)\right]^2. \tag{17}$$

We choose the weights of the first and last layer, respectively W and v, to have Gaussian prior distributions:

$$\mu(W) = \frac{e^{-\frac{\lambda_0}{2}\|W\|^2}}{\mathcal{N}_W}\,,\ \mu(v) = \frac{e^{-\frac{\lambda_1}{2}\|v\|^2}}{\mathcal{N}_v}\,, \tag{18}$$

where $\|\cdot\|$ is the standard Frobenius norm, and the constants $\mathcal{N}_{W/v}$ ensure proper normalization $\int \mathrm{d}\mu(W) = \int \mathrm{d}\mu(v) = 1$. The parameters λ_0, λ_1 can be equivalently thought as Gaussian priors over the weights of each layer, or as L^2 regularization terms rescaled by the temperature T.

A neural network with one FC hidden layer is formally defined starting from the pre-activations at the first layer:

$$h_{i_1}(x) = \frac{1}{\sqrt{N_0}}\sum_{i_0=1}^{N_0} W_{i_1 i_0} x_{i_0} + b_{i_1}\,, \tag{19}$$

where b are the first-layer biases. Since we are interested in regression problems, we add one last readout layer, and we define the function implemented by the one hidden layer FC network as:

$$f_{\text{FC}}(x) = \frac{1}{\sqrt{N_1}}\sum_{i_1=1}^{N_1} v_{i_1}\sigma\left[h_{i_1}(x)\right], \tag{20}$$

where σ is a non-linear activation function. An effective action arises in this setting when one considers the canonical partition function at temperature $T = 1/\beta$ associated to the train loss function $\mathcal{L}$ and the thermodynamic limit $N_1, P \to \infty$ at fixed ratios $\alpha_1 = P/N_1$, and we are able to reduce it to a form suitable to saddle-point integration over two scalar order parameters $\{Q, \bar{Q}\}$, $Z = \int dQd\bar{Q}e^{-N_1 S_{\text{FC}}(\{Q,\bar{Q}\})/2}$. In the case of odd-activation function, the action is given by[38]:

$$\begin{aligned} S_{\text{FC}} &= -Q\bar{Q} + \log(1+Q) \\ &+ \tfrac{\alpha_1}{P}\,\mathrm{Tr}\,\log\beta\left(\tfrac{1}{\beta} + K^{(\text{R})}(X,X;\bar{Q})\right) \\ &+ \tfrac{\alpha_1}{P}\mathbf{y}^T\left(\tfrac{1}{\beta} + K^{(\text{R})}(X,X;\bar{Q})\right)^{-1}\mathbf{y}\,, \end{aligned} \tag{21}$$

where the renormalized kernel $K^{(\text{R})}(X,X;\bar{Q})$ is the $P \times P$ matrix defined in Eq. (13).

The minimization of the effective action in Eq. (21) is straightforward if $\alpha_1 \to 0$, since one easily finds that $Q^* = 0$, $\bar{Q}^* = 1$ and recovers the well-known infinite-width limit NNGP kernel. At finite α_1, one finds data-dependent solutions for the order parameters that produce a renormalization of the infinite-width kernel, as expressed by Eq. (13).

The effective action at proportional width presented in Eq. (21) has been firstly derived for deep linear networks in ref. 32. The non-asymptotic evaluation of the partition function in the linear case is given in terms of Meijer-G functions in ref. 33 for fully-connected networks with single output, and in ref. 53 for FC networks with multiple outputs and CNNs. An effective action for globally gated deep linear networks has also been derived[34]. In ref. 38 we obtained the effective action for one FC hidden layer networks with generic activation function leveraging a Gaussian equivalence[54–58] informally justified via a generalized central limit theorem due to Breuer and Major[59]. A result for L layers can be somewhat bootstrapped from the observation that the stochastic process that describes information flow in shallow non-linear and deep linear networks in the proportional limit is related to a Student's t-process[60] (see also[38] for a more detailed discussion). Recent work considers the proportional setting in a teacher-student scenario, averaging over the data distribution using the replica method[61,62] and a first preliminary rigorous result in an almost-proportional limit[63] appeared very recently, leveraging the interpolation method[64,65].

Shallow architectures with local connectivity: notation and definitions

We consider networks with one hidden convolutional layer or one hidden locally-connected layer followed by a fully-connected readout layer. For simplicity, in this paper we will restrict our analysis to one-dimensional convolutions, but the model can be easily generalized to d-dimensional convolutions. In CNNs, pre-activations in the hidden layer are given by:

$$h_i^a(x) = \frac{1}{\sqrt{M}}\sum_{m=-\lfloor M/2\rfloor}^{\lfloor M/2\rfloor} W_m^a x_{Si+m}\,. \tag{22}$$

Here M is the dimension of the channel mask, S is the stride, the index $i = 1, \ldots, \lfloor N_0/S\rfloor$ runs over the input coordinates, the index $a = 1, \ldots, N_c$ runs over the channels and the index m moves through the spatial mask of the convolutional filter. For simplicity, we define periodic boundary conditions (PBCs) over the input coordinates, and we consider odd values of M. With the addition of a FC readout layer, the function implemented by a CNN is given by:

$$f_{\text{CNN}}(x) = \frac{1}{\sqrt{N_c\lfloor N_0/S\rfloor}}\sum_{i=1}^{\lfloor N_0/S\rfloor}\sum_{a=1}^{N_c} v_i^a\sigma\left[h_i^a(x)\right], \tag{23}$$

where σ is a odd activation function and the v's are the learnable weights of the readout layer. The LCN is obtained adding a i (patch) index to the weight matrix W_m^a in Eq. (22), meaning that there will be a different set of weights for each patch visited by the network.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The CIFAR10[66] and MNIST[67] datasets that we used for all our experiments are publicly available online, respectively at https://www.cs.toronto.edu/~kriz/cifar.html, and http://yann.lecun.com/exdb/mnist/.

Code availability

The Python package[49] to reproduce our experiments with Bayesian networks and compute theoretical predictions from our theory is available on Zenodo https://zenodo.org/records/14216887. The code to reproduce experiments on internal representations with ADAM-

trained networks[51] can be found at the following GitHub link https://github.com/ricaiu/PhD/tree/main/lkr.

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Author contributions

P.R. concieved the original idea. P.R., R.A., and R.P. performed the analytical calculations, supported by A.V. and R.B. Numerical experiments, data analysis and data visualization were carried out by R.A. R.P. and P.B. All the authors contributed to discussing and interpreting the results and to writing and editing the manuscript. R.A. and R.P. contributed equally to the work.

Competing interests
The authors declare no competing interests.

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